JAL NIRIKSHAK

Portable AI-Assisted Field Test Kit for PFAS & Microplastics In Water

Technical Approach | Hardware Architecture | Field Workflow | Feasibility | Digital Monitoring

Project positioning

JAL NIRIKSHAK is a proposed rapid field-screening and surveillance platform. It is designed to identify suspicious samples quickly, record traceable field data, and support laboratory confirmation when a result is critical or outside the validated screening range.

Document Structure

The structure organizes its documentation around overview, problem, proposed architecture, technical approach, technology stack, development status, roadmap, performance notes, impact and research basis.

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1. Project Overview

JAL NIRIKSHAK is a proposed portable field-testing kit for rapid screening and digital monitoring of emerging contaminants in drinking-water sources and water-supply systems. The primary focus is PFAS and microplastics, supported by pH, temperature, electrical conductivity and turbidity measurements.

The system is not intended to replace a laboratory. It creates a field layer between routine sampling and laboratory confirmation: collect a controlled volume, run a guided test, receive a simple screening status, and automatically store time, location, measurements and image evidence.

Important design principle

PFAS and microplastics are not reliably detected by ordinary pH, TDS or turbidity sensors. JAL NIRIKSHAK therefore separates general water-quality sensing from dedicated PFAS and microplastic modules.

2. Problem Statement

Development of portable and user-friendly field test kits for screening emerging contaminants such as PFAS and microplastics in drinking-water sources and water-supply systems, enabling rapid identification of contamination events, timely corrective actions, and digital integration of test results for data-driven monitoring and decision-making.

2.1 Problem in simple language

PFAS and microplastics can be difficult to identify using ordinary field measurements. Detailed analysis can require laboratory instruments, controlled sample preparation and trained analysts. A field team therefore needs a faster screening layer that can be used close to the water source.

2.2 Main challenges

- Laboratory analysis is not convenient for every routine field location.
- Field personnel may not have specialist analytical training.
- Microplastic screening needs controlled filtration and optical analysis.
- PFAS needs a dedicated recognition/sensing mechanism.
- Water matrices can interfere with sensor signals.
- Every result should be linked to sample volume, location and time.
- Critical screening results need confirmatory laboratory analysis.

3. Existing Approaches and Gap

Approach	How it works	Strength	Gap
PFAS laboratory LC-MS/MS	Sample preparation/extraction followed by liquid chromatography and tandem mass spectrometry.	High analytical capability.	Requires laboratory infrastructure and trained personnel.
Microplastic microscopy	Water is filtered and retained particles are inspected or imaged.	Particle counting and morphology.	Manual analysis can be time-consuming; polymer identification is difficult without spectroscopy.
FTIR/Raman	Particles are identified using spectroscopic signatures.	Useful for polymer identification.	Specialized instruments and controlled analysis.
Conventional water sensors	pH, EC, temperature and turbidity are measured electronically.	Fast and field-friendly.	Do not directly identify PFAS or microplastics.
JAL NIRIKSHAK	Combines supporting sensors, controlled sampling, dedicated screening, AI optics and digital records.	Designed for rapid field screening and surveillance.	Needs calibration and validation before high-stakes deployment.

4. Proposed Solution —JAL NIRIKSHAK

JAL NIRIKSHAK is made up of five layers: regulated sample collection, water-quality monitoring, contaminant screening, edge processing, and digital surveillance. The gear is placed in a durable portable box, and the interface reduces operator decisions.

4.1 Core operating principle

• Determine the volume of water to collect.
• Assign sample ID, GPS, and time.
• Measure pH, temperature, EC, turbidity, and optionally free chlorine.
• Filter a specific volume for microplastic screening.
• Use the PFAS screening cartridge/reader.
• Process microplastic pictures with optical and AI approaches.
• Perform quality control checks.
• Provide a straightforward result and suggested action.
• Save raw measurements, pictures, and metadata.

5. Objectives

• Create a portable field screening platform for emerging pollutants.
• Control sampling and filtration to correlate microplastic concentration with volume.
• Integrate water quality sensors.
• Create a PFAS screening interface using a replaceable cartridge design.
• Create an optical microplastic module utilizing filtering, lighting, camera image, and AI-assisted analysis.
• Capture sample ID, GPS coordinates, timestamp, sensor readings, and photographic proof.
• Offer a guided workflow for workers with limited technical knowledge.
• Develop a repeat-testing and laboratory-confirmation pathway.
• Create a scalable digital layer for mapping, trend analysis, and contamination-event detection.

6. Complete System Architecture

End-to-end flow: Water Source -> Sample Collection -> Sample ID/GPS -> Pump and Flow Control -> Water-Quality Sensors -> PFAS Screening + Microplastic Filtration/Optical Screening -> ESP32/Raspberry Pi Processing -> Quality Control -> Local Result -> Dashboard/Cloud Sync -> Alert or Laboratory Confirmation.

Module	Main components	Role
Sample	Bottle, tubing, pump, flow sensor	Controls and measures processed water volume.
Water quality	pH, EC, temperature, turbidity, optional chlorine	Provides contextual measurements.
Microplastic	Filter, optical chamber, camera, UV/blue illumination	Retains and images suspected particles.
PFAS	Replaceable recognition cartridge + reader	Produces a dedicated PFAS screening signal.
Control	ESP32	Reads sensors and controls pump/valves.
AI/vision	Raspberry Pi, OpenCV, lightweight ML	Processes optical images.
Digital	GPS, local storage, Wi-Fi/optional 4G	Stores and transmits traceable records.
User	Touchscreen/OLED, buttons, buzzer	Guides operation and shows results.
Power	Rechargeable battery + protection	Enables field operation.

7. Physical Hardware Architecture

To physically isolate the wet/sample region from electronics, use a sturdy portable box with distinct portions.

• Water-quality bay: pH, EC, temperature and turbidity probes; tubing and cleaning accessories.

• Microplastic bay: pump, flow sensor, filter holder and optical chamber.

• PFAS bay: cartridge storage, reader electronics, and replaceable cartridges.

• Control bay: GPS, display, Raspberry Pi, ESP32, and communications.

• Power bay: protective circuit, charging, and rechargeable battery.

• Consumables bay: gloves, filters, clean sample containers, and reagents as needed.

8. Water-Quality Sensor Layer

These are supporting measurements.

8.1 pH

Gives a general context for the sample and measures acidity and alkalinity. Utilize a serious electrode for a serious electrode prototype.

8.2 Temperature

Keeps track of sample temperature and provides context and correction for other readings. For prototyping, a digital probe that is waterproof is appropriate.

8.3 EC/Conductivity

Measures electrical conductivity and can indicate changes in dissolved ionic content. It cannot prove PFAS contamination.

8.4 Turbidity

Measures cloudiness/suspended material. It helps assess sample condition but is not a microplastic counter.

8.5 Free chlorine

Useful for treated drinking-water monitoring. A controlled colorimetric chamber can measure reagent color using the camera.

Rule

pH + EC + temperature + turbidity are contextual sensors. PFAS and microplastics need dedicated detection modules.

9. Microplastic Detection Module

The microplastic subsystem uses controlled-volume filtration followed by optical imaging. The intended output is a screening estimate such as suspected particles per litre, supported by image evidence.

9.1 Step-by-step working

1. Collect the sample and connect the tube.
2. Fill the filter with a specific volume, such as 1 liter.
3. Set the flow sensor to stop at the desired volume.
4. Place the filter in the optical chamber.
5. For fluorescence-based methods, apply regulated UV/blue lighting.
6. Take photos with specified geometry and exposure.
7. Preprocess images to remove background noise.
8. Detect candidate particles using AI/vision.
9. Provide an estimate for count, size, and features, along with a confidence value.
10. Report suspicious results for further testing and lab confirmation.

9.2 Why flow measurement matters

If 1.0 L is processed and 42 candidate particles are detected, the screening result can be expressed as approximately 42 suspected particles/L. Reliable volume measurement is therefore essential for sample-to-sample comparison.

9.3 Limitations

- Optical detection can produce false positives.
- Very small particles may be difficult to resolve with low-cost optics.
- Fluorescent staining is not automatically definitive polymer identification.
- Polymer identity should be confirmed using an appropriate validated method when required.

10. PFAS Screening Module

PFAS is the most demanding part of the proposed kit. The system should use a dedicated sensing cartridge rather than inferring PFAS from pH, TDS, turbidity or conductivity.

10.1 Proposed architecture

Water sample -> controlled preparation/recognition layer -> PFAS-sensitive interaction -> electrochemical or optical signal -> reader electronics -> calibrated screening result.

10.2 Research directions

- Molecularly imprinted polymer recognition layers.
- Electrochemical sensing with functionalized electrodes.
- Optical recognition approaches.
- Replaceable cartridge design for maintenance and contamination control.

10.3 Result logic

The interface can show GREEN — no indication detected under the validated condition; YELLOW — suspected/near threshold and repeat recommended; RED — positive screening requiring sample preservation and laboratory confirmation.

Laboratory confirmation

Validated laboratory approaches such as SPE followed by LC-MS/MS are reference approaches for PFAS analysis. JAL NIRIKSHAK positioned as a field-screening layer, not a replacement for validated laboratory quantification.

11. Embedded Control and AI Processing

11.1 ESP32 responsibilities

• Monitor pH, EC, temperature, turbidity, and flow signals.
• Control the pump and optional valves.
• Monitor pressure and irregular flow.
• Manage device state and error conditions.
• Communicate with Raspberry Pi.
• Provide alerts for low battery, blockage, and sensor errors.

11.2 Raspberry Pi responsibilities

- Run camera capture and optical analysis.
- Perform OpenCV preprocessing.
- Run lightweight object detection/segmentation.
- Store local images and records.
- Generate the result screen.
- Synchronize records when connectivity is available.

11.3 Software stack

Layer	Suggested technology
Embedded	ESP32 / Arduino framework or ESP-IDF
Edge AI	Python, OpenCV, PyTorch or ONNX Runtime
AI model	Lightweight YOLO/object detection or segmentation
Local data	SQLite / JSON
Backend	FastAPI
Database	PostgreSQL; PostGIS for location analytics
Frontend	React.js + Tailwind CSS
Maps	Leaflet.js
Communication	Wi-Fi; optional 4G; Bluetooth
Reporting	PDF/CSV export

12. Digital Monitoring and Dashboard

The digital layer converts individual field tests into a surveillance system. Every test should have a unique sample ID and minimum metadata.

- Sample ID, date and time.
- GPS/location and operator/device ID.
- pH, EC, temperature, turbidity and optional chlorine.
- Processed water volume.
- PFAS screening status.
- Suspected microplastic count and confidence.
- Raw image evidence.
- Calibration/QC status.

- Final action: normal, repeat, investigate or laboratory confirmation.

12.1 Dashboard features

- Sampling-location map.
- Green/yellow/red status.
- Time-series trends.
- Sample history and raw evidence.
- Calibration and battery status.
- Filter/cartridge reminders.
- Exportable reports.
- Laboratory-confirmation tracking.

13. User Workflow for Non-Expert Personnel

The operator should not have to comprehend the chemistry or AI model. The device should guide the user through a predetermined sequence.

1. Power on and do automatic self-checks.
2. Scan or create a sample ID.
3. Follow onscreen directions to collect the sample.
4. Connect sample and tubing.
5. Press the START button, and the pump will automatically control the volume.
6. Wait for sensors and optical analysis to run.
7. Read the simple status first, followed by detailed measurements.
8. Take recommended action: log, repeat, investigate, or confirm in laboratory.
9. Sync to the dashboard when internet is accessible; otherwise, store locally and sync later

14. Reliability, Quality Control and Validation

Control	Implementation	Purpose
Known volume	Pump + calibrated flow sensor	Improves concentration calculations.
Blank test	Clean-water/filter blank	Detects background contamination.
Calibration	Known standards/reference materials	Converts raw signals into measurements.
Duplicate test	Repeat suspicious samples	Checks reproducibility.
Image evidence	Save raw and processed images	Makes AI output auditable.
Confidence score	Model confidence + QC state	Avoids overconfident interpretation.
Sensor diagnostics	Range/drift checks	Detects faulty probes.
Lab comparison	Compare with reference results	Measures field-kit performance.

14.1 Validation plan

1. Bench calibration using clean water and known standards.
2. Interference testing with different water matrices.
3. Repeatability testing.
4. Controlled microplastic samples with known particle characteristics.
5. PFAS reference samples across the intended screening range.
6. Blind comparison against laboratory results.
7. Field pilot across multiple water sources.
8. Statistical analysis of false positives, false negatives and repeatability.

15. Feasibility and Viability

15.1 Technical feasibility

Core sensors, embedded controllers, cameras, pumps and displays are commercially available. The main research challenges are the dedicated PFAS module and validation of the microplastic optical workflow. A staged prototype is therefore practical.

15.2 Operational feasibility

- Battery operation for field locations.
- Offline-first storage where internet is unavailable.
- Replaceable filters and cartridges.
- Color-coded connectors and guided prompts.
- Automatic volume control.

15.3 Deployment feasibility

The platform can be modular: low-cost core sensors support routine measurements while dedicated contaminant cartridges are replaceable consumables. Multiple units can feed a central dashboard.

16. Innovation and Uniqueness

- Dual emerging-contaminant screening in one portable platform.
- Controlled water volume for microplastic concentration estimates.
- AI-assisted optical screening with image evidence.
- Replaceable PFAS cartridge concept.
- GPS, timestamp and sample ID for traceability.
- Offline-first operation.
- Automatic QC and repeat-test recommendations.
- Simple workflow for non-expert personnel.
- Direct pathway from screening to laboratory confirmation.

17. Impact and Benefits

Area	Expected contribution
Public health	Earlier identification of suspicious water-quality events can support faster investigation.
Rapid response	Field screening provides an initial indication closer to the source.
Surveillance	Portable units can expand sampling coverage.
Data-driven monitoring	GPS-tagged records enable trends and maps.
Operational efficiency	Suspicious samples can be prioritized for laboratory testing.
Accessibility	Guided workflow lowers technical barriers.
Traceability	Measurements, images and actions remain linked to the sample.

18. Development Status

The project should clearly separate implemented features, prototype-ready modules and research modules. This prevents design targets from being presented as measured results.

Stage	Content
Defined	Architecture, sensor list, hardware concept, workflow and data model.
Prototype-ready	ESP32 sensing, temperature/EC/pH/turbidity, pump and flow control, logging.
In development	Microplastic filtration, optical chamber, image processing and AI.
Research/validation	PFAS cartridge/reader, calibration and laboratory comparison.
Future integration	Cloud dashboard, multi-device monitoring and field pilot.

19. Development Roadmap

9. Requirements and measurement workflow.
10. ESP32 sensor and pump-control prototype.
11. Sample-volume control and filtration.
12. Optical chamber and camera pipeline.
13. Microplastic dataset collection and labeling.
14. AI model training and validation.
15. PFAS cartridge/reader research prototype.
16. Database and dashboard.
17. Complete hardware integration.
18. Calibration, blanks, duplicates and interference testing.
19. Laboratory comparison.
20. Field pilot and usability evaluation.

20. Performance and Validation Notes

Do not claim guaranteed detection limits, regulatory compliance or laboratory-equivalent accuracy until these values are experimentally established.

- Detection limit: report only after calibration/validation.
- Accuracy: compare with a reference method.
- Precision: report repeated-measurement variation.
- False-positive and false-negative rates: measure with known samples.
- AI performance: report precision, recall and F1-score for microplastic detection.
- Field reliability: measure battery life, sensor drift, filter failures and operator errors.
- End-to-end latency: measure sample insertion to final result.

21. Future Scope

- Multi-PFAS sensor arrays.
- Improved recognition materials.
- Smaller-particle optical detection.
- FTIR/Raman-assisted polymer confirmation.
- Larger AI datasets across water matrices.
- Automated cartridge identification and expiry tracking.
- 4G/5G or LoRaWAN distributed monitoring.
- Regional contamination maps and anomaly detection.
- Laboratory information-system integration.
- Predictive analytics from historical water-quality trends.

22. Research Basis and References

U.S. EPA — PFAS Drinking Water Laboratory Methods — https://www.epa.gov/pfas/epa-pfas-drinking-water-laboratory-methods

PFAS drinking-water analytical methods, including Methods 533 and 537.1.

WHO — Microplastics in Drinking-water — https://www.who.int/publications/i/item/9789241516198

Background on microplastics in drinking water, monitoring and research needs.

ISO 5667-27:2025 — https://www.iso.org/standard/87475.html

Guidance on sampling for microplastics in water.